Course Code: PRJ1101	**Course Name:** Joy of Engineering	
Credits: 3 (2-0-2)	**Contact hours per week:** 2 hours Theory & 2 hours lab per week	
Batch: 2024, 2nd Sem **Academic Year:** 2024-25	**Semester Duration:** Jan - May 2025	
Course Coordinator – **Name: Hirdesh Kumar Pharasi** **Email:** hirdesh.pharasi@bmu.edu.in **Office:** Room no. 89, fourth floor, E2 Building, BMU **Phone:** 9775550708		

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Aim of the course: This course is designed to enable participants to:

- Encourage students to develop solutions for socio-economic challenges, focusing on practical problem-solving skills.
- Cultivate critical thinking, entrepreneurial abilities, and ethical leadership by combining technical expertise with essential life skills for real-world problem-solving.
- Facilitate collaborative learning through group projects that enhance teamwork and project management skills.
- Introduce emerging technologies, providing insights into their future impact on engineering.
- Offer a platform for developing small hardware and software projects, encouraging hands-on learning.

Course Overview and Context: The current start-up ecosystem inspires young minds to tackle societal challenges through innovation, and the Joy of Engineering (JOE) course provides a similar environment. In this course, students gain foundational knowledge essential for designing and completing projects within a set timeframe. They receive guidance in developing product prototypes and exploring key concepts across various areas, including Project Management, Single-board Computing, Web Development, Mobile App Development, Programming, IoT and Sensors, Business Incubation, and Applications of Artificial Intelligence and Machine Learning.

Course Outcomes: At the end of the course, students should be able to

CO1: Understand various technical concepts and identify a suitable/feasible problem for the project

CO2: Analyze multiple possible solutions and technical specifications for the project.

CO3: Create a prototype of the project with the required hardware/software simulation results.

Course Competencies: (Course Outcomes further elaborated) and Instruction Schedule:

Topics	**No. of Sessions & CO**	**Faculty Members**	**Topic Description**
A bridge between Design thinking and JOE	1 & CO1	Dr. Hirdesh	Design Thinking and the Joy of Engineering (JOE) share a common foundation of creativity, problem-solving, and hands-on learning. Design Thinking emphasizes user-centric solutions and JOE fosters curiosity and practical engagement with emerging technologies, enabling students to develop teamwork, communication, and lifelong learning skills.
Problem Understanding/ Feasibility	4 & CO1	Faculty mentors	Faculty mentors will evaluate project feasibility during the Design Thinking course.
Single board computing and related programming concepts	2 & CO2, CO3	Dr. Soharab/ Dr. Sandeep	The students will understand the architectures of several microcontrollers and learn programming skills to use them for various applications. Most of the work will be based on hands-on practice, encompassing qualities such as teamwork, communication, collaboration, and decision-making among the students.

IoT & Sensor-based Applications	4 & CO2, CO3	Dr. Sandeep/ Dr. Pooja	Students will learn the fundamentals of IoT systems, sensor technologies, and communication protocols. They will gain hands-on experience in designing and prototyping IoT-based applications, integrating sensors with platforms like Arduino and exploring real-world use cases in smart homes and healthcare etc.
Electrical circuits and hardware, PCB fabrication	2 & CO2, CO3	Dr. Anubhav/ Dr. Pooja	Students will learn to design and analyze electrical circuits, understand hardware components, and create schematic diagrams. They will gain hands-on experience in PCB fabrication, including layout design, soldering, and assembling functional electronic hardware.
Web Development Concepts	2 & CO2, CO3	Dr. Kiran Khatter	The sessions will focus on building structured and visually appealing web pages while emphasizing lifelong learning, communication, and teamwork in collaborative development projects
Mobile Application Development (MAD)	2 & CO2, CO3	Dr. Nikhil	The main focus of MAD sessions will be to develop Android Applications for Smartphones. These sessions provide hands-on experience and guidance, enabling participants to learn the essential basic skills for building Android apps. This will also foster connections with other developers, share ideas with the team collaborators.
AI/ML/NLP Applications	2 & CO2, CO3	Dr. Shakti/ Dr. Atul	The sessions will focus on the foundations of AI/ML, covering core principles and algorithms, the role of data, and key machine learning techniques. Participants will gain both theoretical understanding and practical insights into real-world applications.
Digital Image processing	2 & CO2, CO3	Dr. Devanjali	Digital Image Processing (DIP) involves manipulating images using algorithms to enhance or extract useful information. Students start with basic operations like reading images, image filtering, various pre-processing techniques, Overview on basic segmentation techniques, etc.
Solid Modeling and Simulation	2 & CO2, CO3	Dr. Srikanth/ Dr. Yogesh	The students will be introduced to 1D through 3D modeling practices, an indispensable aspect in the design of the prototype. Furthermore, they will be introduced to simulation tools to assess the feasibility and arrive at an optimal design whilst meeting the required constraints.
Technical Report Writing	2 & CO2	Dr. Sridharbabu/ Dr. Vishal	Technical report writing is a structured approach to documenting and conveying technical information effectively. It serves as a vital communication tool in professional and academic environments, emphasizing clarity, precision, and objectivity.

Business Incubation Concepts	2 & CO2, CO3	Dr. Deepak	The session will give the students a high-level view of the various concepts involved in venture creation and taking it up to a start-up. It will talk about the various types of funding opportunities available and will act as a mirror for the students to assess their current preparedness as entrepreneurs. Venture Creation Process , Pre Incubation , Seed , Product-Market fit , Stages of Funding , Prototype , MVP , Angel, VC , Seed Fund , Selecting an Incubator , Startup vs Business vs. venture MVs founder.
Project Management	1 & CO1, CO2	Dr. Amiya/ Dr. Ranbir	Students will learn the fundamentals of project planning, including how to define clear objectives, establish the scope, outline key deliverables, set a structured timeline, allocate budgets, and apply Critical Path Analysis (CPA) to identify essential tasks.
Hands-on	32 & CO3	Faculty Mentors	Students will focus on developing their JOE prototype.

Total Hours for Course: 64 = 32 (Th) + 32 (Hands-on)

Topics of the Course

- Introduction to the course.
- Briefing on various project ideas and project topic finalization after feasibility analysis – preparation of the Project Abstract
- Introduction to Project Management
- Single board computing and related programming concepts
- Introduction to IoT and Sensors,
- Web and Mobile development concepts,
- Applications of Artificial Intelligence and Machine Learning
- Refinement and finalization of the scope of the Projects - objectives, deliverables followed by Synopsis Presentation
- Project development/simulation and Mid Semester Presentation
- Submission of the final Project Report and End Semester Presentation

The JOE course represents the broad areas, where students will work on at least one project of their choice (the project will be the same selected during the course “Design Thinking - The Way to Joy of Engineering” or floated by faculty or proposed by students themselves and validated by faculty for scope and feasibility). Students must ensure that their final choice of project is a demonstrable artifact like a working prototype (**hardware**) of a product and a working software solution like a mobile app/Web app. The technology may

include the use of sensors, IoT, embedded vision-based automation, and computer programming for developing smart or intelligent applications.

CO-PO Mapping

1-Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation (CSE)

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
↓ CO Mapping Course Mapping→																
CO1	1		1	3					2			2	2			
CO2		1			3	3		3	2	3	2	1	2	2		
CO3					3					3	3	3		3		

1-Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation (EComE)

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
↓ CO Mapping Course Mapping→																
CO1	1		1	3								2	1			
CO2		1	2		3	3		3	2	3	2	1	3	2		
CO3			3							3	3	3		3		

1-Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation (ME)

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
↓ CO Mapping Course Mapping→																
CO1	1		1	3								2	1			
CO2		1	2		3	3		3	2	3	2	1	3	2		
CO3			3							3	3	3		3		

Note: For details about PO, PSO, and CO-PO mapping, refer to the OBE document for B. Tech. – CSE

Learning Resources:

Reference Books:

1. Alley, Michael, The Craft of Scientific Writing, Springer New York, (1996).
2. Saikat Dutt, Subramanian Chandramouli, and Amit Kumar Das, Machine Learning, Pearson, 2018.
3. Introduction to solid modeling using Solidworks, McGraw Hills, 2024.
4. Lecture notes and hands-on workshop: A gentle introduction to finite element analysis.

5. Rajkumar Buyya, and Amir Vahid Dastjerdi, Internet of Things: Principles and paradigms. Elsevier, 2016.
6. Raj Kamal, Internet of Things, McGraw Hill, 2017.
7. John Horton, Android Programming for Beginners, Third Edition, Packt Publishing, 2021.
8. Gonzalez, Rafael C. Digital image processing. Pearson education india, 2009.
9. Oskay, Windell, and Eric Schlaepfer. Open Circuits: The Inner Beauty of Electronic Components. No Starch Press, 2022.

Reference Links:
https://developer.mozilla.org/en-US/docs/Web/HTML
https://developer.mozilla.org/en-US/docs/Web/CSS

E-Learning (and other) Resources: Other relevant study material, MOOCs (and other e-learning material) will be suggested and discussed during the lecture session(s) as and when deemed suitable.

Assessment Pattern:
The final grade will be based on the marks/ grades obtained in the end-semester exam and other assessments defined in the assessment table below. The relative grading method described in the university's academic regulations will be followed to grade the students.

Sessional:

Evaluation Component	Weightage (60%)	Evaluation Schedule	Rubrics/Remarks
Quiz	20%	Mid-Term Week	Based on the various topics to be discussed in the class.
Presentation	40%	The first week of April	During the presentation, students must present their project progress as part of the evaluation, followed by a viva. Additionally, they must submit a short report by 26th March (a template will be provided).

End term Examination:

Evaluation Component	Weightage (40%)	Evaluation Schedule	Rubrics/Remarks
Final Report	15%	30th April, 2025	Project report (10%) + Poster submission (5%)
End-Term-Presentation	25%	7th May, 2025	Concept implementation and demonstration of the working prototype/simulation during JOE Exhibition

Experiential Learning: This course contains above 50% experiential learning components.

Student Responsibilities:

- Regular and attentive during lectures/lab classes.
- Write class notes and submit assignments on time.
- Check announcements at Maitri / Google Classroom / E-mail regularly.
- Regularly check marks and attendance on the Maitri/Google Classroom and inform if there are any concerns.

Support for Slow Learners:

- Extra classes as per the requirements.
- Additional assignments.
- Group them with advanced learners for completing assignments / Lab sessions.

Challenges for Advanced Learners:

- Providing higher level numerical problems / real life applications through proper guidance.

Attendance Policy: As per the University policy.

Quiz / Assignments: One Quiz and One Assignment as per the schedule provided.

Evaluation procedures for tests, assignments, and lab: As per the weightage mentioned above.

Late assignment submission policy: Students will be awarded zero for late submission except for the situations like a medical problem.

Make-up examination/work policy: As per the University policy

Behaviour expectations: Students are advised to attend classes regularly and be punctual. The use of mobile phones during lectures and lab work is strictly prohibited. Ensure academic integrity – no copying, no cheating, focus on learning and gaining knowledge.

Academic dishonesty/cheating/plagiarism: Any student violating expected behaviour will be punished according to the University's rules and regulations.